

# Diksha Damahe

AI/ML Engineer | +91-9770925370 | [dikshadamahe1@gmail.com](mailto:dikshadamahe1@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## Education

### VIT Bhopal University

2023–2027

*B.Tech Computer Science & Engineering (Artificial Intelligence and Machine Learning) | CGPA 8.03*

## Technical Skills

**Languages:** Python · Dart · TypeScript · SQL

**Machine Learning:** scikit-learn · XGBoost · LightGBM · CatBoost · Isolation Forest · SMOTE · SHAP · SciPy

**NLP / DL:** Sentence-Transformers (BERT) · TensorFlow · Groq / Llama 3 tool-calling

**Serving & data:** FastAPI · Django REST · Pandas · NumPy · PostgreSQL · SQLite · MongoDB

**Clients:** Flutter · React · Next.js · PyQt5

**Tools:** Git · GitHub Actions · Vercel · Render · AWS · Jupyter

## Experience

### Hindustan Copper Limited (Malanjkhand Copper Project)

May 2026 – Jul 2026

*AI/ML Engineering Intern*

- Built **RecovAI** on **1,778** digitized shift logs and **26** process variables; **XGBoost** recovery  $R^2$  **0.972** / RMSE 0.139 / MAE 0.104.
- Isolation Forest anomalies, **SHAP**, PSI drift, SciPy dose search; **FastAPI** + Groq Llama 3 ReAct agent. Live: [hcl-recovai.onrender.com](https://hcl-recovai.onrender.com).

## Projects

### RecovAI — Industrial ML Decision Support

May 2026 – Jul 2026

*XGBoost · SHAP · Isolation Forest · FastAPI · Groq · [Live](#) · [GitHub](#)*

- Plant-facing recovery predictor, reagent optimizer, anomaly flags, and auto shift reports for Hindustan Copper.

### NAFLD Risk Stratification Pipeline

Feb 2026 – Mar 2026

*scikit-learn · XGBoost / LightGBM / CatBoost · SMOTE · SHAP · [GitHub](#)*

- Merged **6 NHANES 2017–18** files; leakage-safe SMOTE on the training fold only. Benchmarked **24 classifiers**; Random Forest **0.9644 test ROC-AUC**, 84% sensitivity, 91% specificity.

### SPARTA — Phishing Website Detection

2025 – 2026

*Python · scikit-learn · IEEE SCEECS 2026 · [DOI](#)*

- Ensemble detector on lexical and host-based URL features; **99.15%** detection on PhishTank and UCI.

### Veil Android

Apr 2026 – Present

*Flutter · Dart · ExoPlayer · Riverpod · Hive · [GitHub](#)*

- Production Flutter client for Android phone, tablet, and TV: TMDB catalog, REST resolver, ExoPlayer HLS/DASH, Hive persistence, signed CI/CD releases (**v3.1.1**).

### QuantFactor — Factor Research & Backtesting

Aug 2026

*Python · pandas · NumPy · SciPy · [GitHub](#)*

- Point-in-time backtester with  $t+1$  execution lag, commission/spread/slippage, Bonferroni and Benjamini–Hochberg FDR, walk-forward OOS, VaR / CVaR.

### Note Assistant — BERT Semantic Search

Mar 2026

*Python · Sentence-Transformers · all-MiniLM-L6-v2 · [GitHub](#)*

- CLI over personal notes: BERT embeddings, cosine ranking, on-disk cache, 30-question precision / recall / F1 harness.

## Publications

**SPARTA:** Secure Phishing Analysis and Robust Training Algorithm for Phishing Website Detection. Diksha Damahe, Ajay Kumar Phulre. *IEEE SCEECS 2026* (Scopus). DOI: [10.1109/SCEECS68810.2026.11430134](https://doi.org/10.1109/SCEECS68810.2026.11430134)

## Certifications

Applied Machine Learning in Python — University of Michigan (Nov 2025) · Google IT Support — Google (Jan 2026) · AWS Technical Essentials — Amazon Web Services (Sep 2025) · Cloud Computing — NPTEL, IIT Kharagpur (May 2025) · Introduction to Internet of Things — NPTEL, IIT Kharagpur (May 2026)

## Hackathons

**Adobe University Hackathon 2026** — Aug 2026 · **Women Who Master Hackathon 2026** (Aspire For Her × Logitech) — Jul–Aug 2026 · **ET AI Hackathon 2.0** (The Economic Times) — Jun–Jul 2026 · **FlowZint AI Hackathon 2026** — May–Jul 2026